

[54] ALUMINUM ALLOYS
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[57] ABSTRACT
The aluminum alloy AA 6063 containing 0.2–0.6% Si and 0.45–0.9% Mg is widely used for the production of extrusions. By incorporating 0.02–0.3% Mn or 0.02–0.3% Mo in AA 6063 alloy in which the Fe content is controlled to lie in the range of 0.1–0.25% Fe and is related to the Si content in excess of that required to convert all the Mg content to Mg₂Si, it is found that significant improvements in the extrusion properties of the metal are obtained by converting the Al-Fe-Si phase to cubic α -phase by heating the ingot at a temperature of 550°C for about 1 hour.

5 Claims, No Drawings

The present invention relates to aluminium alloys and in particular to aluminium alloys containing magnesium and silicon. An alloy containing 0.45 – 0.9% Mg and 0.2 – 0.6% Si is the most widely used alloy for the production of aluminium extrusions. This alloy is widely known under the United States Aluminum Association Standards as Alloy 6063. The specification of Alloy 6063 states maximum permissible levels of various impurity metals and inter alia permits the inclusion of Mn in amounts up to 0.1%. The specification permits the presence of Cu, Cr, Zn and Ti up to level of 0.1% and also other impurities in minor amounts up to a total of 0.15% and 0.05% each, whilst Fe is permissible in amounts up to 0.35%.

In the as-cast condition the alloying elements and the impurities present in the extrusion ingot are either in solid solution in the aluminium matrix or segregated in the form of intermetallic phases at the boundaries of the grains into which the alloy has solidified, or at the boundaries of the dendrite cells within those grains. It has been common practice for some years to homogenise the structure of 6063 alloy by a heat treatment aimed at eliminating coarse particles of the magnesium silicide phase and the micro-segregation or 'coring' of magnesium and silicon in the dendrite cells, since it is not possible to obtain the optimum properties in extrusions produced from ingots containing such segregation of magnesium and silicon. In one widely used practice the as-cast 6063 ingots are heated for several hours at a temperature of about 550°C. and cooled rapidly so as to lock a large proportion of the magnesium silicide in solution and to ensure that the remainder of this phase is precipitated in the form of very finely dispersed particles which can be readily dissolved by re-heating to homogenising temperature. Extrusions with very good mechanical properties can be produced at favourable extrusion pressure and speeds if ingots homogenised in this way are re-heated quickly to a temperature below the homogenising temperature immediately prior to extrusion. However the surface finish of extrusions produced from such ingots is not always as good as is desired.

The surface finish of an extrusion is to a large extent dependent on the speed at which the metal is forced past the die. It is an object of the present invention to provide an improvement of the Alloy 6063 composition, which permits an increase in the extrusion rate without loss of surface finish of the extrusion or conversely to provide a better standard of surface finish without change of extrusion rate as compared with a typical specimen of Alloy 6063 coming within the general specification.

A principal cause of defects in the quality of the surface finish of Alloy 6063 extrusions is the breaking away of solid components from the surface of the metal as it is forced through the die orifice. These defects manifest themselves as light-coloured micro score-lines on the surface of the extrusion and are commonly referred to as 'pick-up'. The incidence of pick-up and its deleterious effect on the reflectivity and apparent colour of the surface of the extruded metal increase with the extrusion speed.

It has now been postulated that one of the principal causes of surface defects from pick-up is due to the presence of the intermetallic phase called β -Al-Fe-Si in

the ingot during the extrusion process. This phase, which is insoluble under the normal homogenisation conditions referred to above, grows in the form of thin, brittle sheets and is formed during the production of the ingot by the direct chill casting process. It is believed to have the chemical formula $\text{Fe}_2\text{Si}_2\text{Al}_9$ and has a monoclinic type crystal structure. The insoluble Fe-rich phase can also be present in a different form, α -Al-Fe-Si. This phase is believed to have a chemical formula $\text{Fe}_3\text{Si-Al}_{12}$ and has a cubic type crystal structure. It has now been found that substantial reduction in pick-up defects can be achieved if the β -Al-Fe-Si phase present in the as-cast ingot can be converted to the α -phase before extrusion of the ingot, since it is less mechanically brittle than the β -form. This is particularly true whilst the level of iron in the alloy is maintained within the range of 0.1 – 0.25%. Above 0.25% Fe pick-up tends to increase, irrespective of the phase of the aluminium-iron-silicon intermetallic compound, whilst below 0.1% Fe it becomes difficult to obtain the common uniform matte etched finish with existing commercial etch solutions.

It has been realised that the β -phase is a meta-stable phase which can slowly be converted into the α -phase or the FeAl_3 phase by prolonged heating of the ingot but this is not practicable for economic reasons, although the effect of the form of the Al-Fe-Si phase on the extrusion characteristics of a 6063 alloy ingot had not previously been understood. The present invention relies on formulating an alloy of the 6063 type with which it is possible to obtain conversion of substantially the whole of the β -Al-Fe-Si phase into the α -phase under economically-practicable homogenisation conditions; that is to say, within a relatively short time.

To achieve this result under practical conditions a composition must be formulated which provides reproducible results under practical conditions; that is to say, the alloy composition required to obtain desired results must not be so rigid that it is impracticable to follow it under normal commercial melting procedures.

It has been realised that most, if not all, of the additional components permissible in Alloy 6063 will have some effect on the transition of β -Al-Fe-Si to the α -form, either to promote or to hinder the transition. It has now been found that the presence of manganese in amounts of 0.02 – 0.3% in aluminium-magnesium-silicon alloys of the type under discussion promotes the transition of β -Al-Fe-Si to α -Al-Fe-Si. It is also found that the transition is promoted by the presence of molybdenum in amounts of 0.02 – 0.3%. The transition from β -Al-Fe-Si to α -Al-Fe-Si in the AA 6063 alloy may be achieved under economically acceptable heat treatment conditions (approximately the conditions required for removal of the coring of magnesium and silicon) when the atomic ratio of the Si content, in excess of the stoichiometric amount required to convert all Mg present to Mg_2Si , to the Fe content (considered as 0.25% Fe) lies within the range 0.7 – 2.5:1. Preferably the normal impurity contents of aluminium are held below 0.2%, the contents of Cu and Zn each being held below 0.03% and the content of Cr below 0.01%.

Stated in another way, it is found that alloys lying within the AA 6063 specification are found to provide extrusions having good surface characteristics when extruded at high rate when the alloy contains 0.02 – 0.3% Mn or Mo, 0.1 – 0.25% Fe and the silicon content exceeds that required to convert the whole of the magne-

sium to Mg_2Si by an amount of 0.09 – 0.32%. Preferably, the copper, chromium and zinc impurities are held within the levels above stated. When Mo is incorporated in the alloy, preferably no deliberate addition of manganese would be made, so that the manganese level would be at the normal impurity level of commercial purity aluminium, generally below 0.01%.

Aluminium-magnesium-silicon alloys of the above composition are found to have improved surface finish as compared with extrusions formed from the same alloy without added manganese or molybdenum but homogenised and extruded under exactly the same conditions.

Whilst the addition of manganese or molybdenum in the stated amounts is helpful in the improvement in the surface characteristics of extrusions throughout the range of the AA 6063 alloy, it is preferred to hold the magnesium content below 0.7%.

Whilst, as stated above, addition of manganese or molybdenum in amounts of 0.02 to 0.3% is envisaged, most of the benefits of the invention are obtained by incorporating manganese or molybdenum in an amount of about 0.05 to 0.1%. With alloys incorporating Mn in amounts in excess of 0.05% transformation of β -Al-Fe-Si to the α -phase may be achieved at 550°C in about 1 hour, whilst with lower manganese contents either a longer heating period or a higher temperature is required. Thus it is always preferred to incorporate Mn or Mo in amounts in excess of 0.05%, the preferred amount being 0.05 – 0.1%.

One alloy according to the invention had the following composition: Si 0.40 – 0.47%; Mg 0.47 – 0.53%; Fe 0.17 – 0.25%; Mn 0.05 – 0.10%; total other impurities 0.2% (max.) (the contents of Zn, Cr and Cu being held within the above stated preferred maxima); Al balance. When this material was extruded it was found that there was a significant improvement in the specular reflectivity and whiteness of the extrusion as compared with the extrusions of the same alloy, but having a normal manganese content of about 0.01%, and extruded under the same conditions.

In one series of tests carried out with this alloy the manganese level was held at the following levels (a) below 0.01%, (b) 0.1%. The ingots were homogenised at 550°C for 1 hour to homogenise the magnesium silicide and were extruded, after normal reheating at 225 ft./min., with the following results:

	a	b
Specular Reflectivity	14%	21%
* Image Clarity	27	41
* Whiteness	78	70
Visual Comparison	Very Heavy Pick-up	Light Pick-up

* Measured by methods described by B.W. Robinson in "Metal Finishing", February 1970.

In another series of tests the time required to transform the β -phase to the α -phase was measured for different Mn levels at 550°C.

% Mn	
>0.01	more than 24 hrs.
0.02	7 hrs.
0.04	2 hrs.
0.06	¾ hr.

-Continued

% Mn	½ hr. 20 mins.
0.07	
0.10	

From the above examples it will be seen that in the case of manganese, an addition of at least 0.05% Mn is to be preferred, because substantially complete transformation of the β -phase to the α -phase may be achieved within the time and temperature conditions normally employed for the homogenisation of the AA 6063 alloy.

In a further series of tests extrusions were produced from ingots under the following conditions:

Ingot:	A	0.48% Mg, 0.46% Si, 0.13% Fe:	balance Al
	B	do.	plus 0.08% Mn
	C	0.46% Mg, 0.45% Si, 0.20% Fe:	balance Al
	D	do.	plus 0.08% Mn

Conditions: Ingot homogenised as tabulated below, preheated to 425°C, and extruded at 250 ft/min.

— Ingot homogenised at 555°C for about 1 hour				
	A	B	C	D
Specular Reflectivity	17%	52%	18%	47%
Image clarity	26	32	31	30
Whiteness	89	56	82	61
Visual comparison	Heavy pick-up	Nil pick-up	Heavy pick-up	Nil pick-up

II — Ingot homogenised at 580°C for about 1 hour				
	A	B	C	D
Specular Reflectivity	15%	39%	14%	37%
Image clarity	27	31	31	31
Whiteness	85	67	84	65
Visual comparison	Heavy pick-up	Nil pick-up	Heavy pick-up	Nil pick-up

- We claim:
1. An aluminium magnesium silicon alloy consisting essentially of 0.2 – 0.6% Si and 0.45 – 0.9% Mg, 0.02 – 0.3% of a metal selected from the group consisting of Mn and Mo, 0.1 – 0.25% Fe, the amount of Si in excess of that required to convert all Mg to Mg_2Si being 0.09 – 0.32%, less than 0.1% Cu, less than 0.1% Zn, less than 0.1% Cr, less than 0.1% Ti, less than 0.15% other impurities, balance aluminium.
 2. An aluminium magnesium silicon alloy according to claim 1 in which the manganese content is 0.05 – 0.1%.
 3. An aluminium magnesium silicon alloy according to claim 1 characterized in that the molybdenum content is 0.05 – 0.1%.
 4. An aluminium magnesium silicon alloy according to claim 1 consisting essentially of Si 0.40 – 0.47%, Mg 0.47 – 0.53%, Fe 0.17 – 0.25%, Mn 0.05 – 0.5%, less than 0.03% Cu, less than 0.03% Zn, less than 0.01% Cr, total impurities (including Cu, Zn and Cr) being less than 0.2%, balance aluminium.
 5. An aluminium magnesium silicon alloy according to claim 1 in which the impurity levels for copper, zinc and chromium are held below the following maxima: Cu 0.03%, Zn 0.03% and Cr 0.01%.

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UNITED STATES PATENT OFFICE
CERTIFICATE OF CORRECTION

Patent No. 3,879,194 Dated April 22, 1975

Inventor(s) Larry Roy Morris et al.

It is certified that error appears in the above-identified patent and that said Letters Patent are hereby corrected as shown below:

Column 4, line 25, in the table heading, -- I -- should appear before "Ingot", and "555° C" should read -- 550° C --; line 26, heading "B" should be moved to the left, so as to appear over the column beginning "52%"; and -- C -- should appear over the column beginning "18%".

Signed and Sealed this

Sixteenth **Day of** November 1976

[SEAL]

Attest:

RUTH C. MASON
Attesting Officer

C. MARSHALL DANN
Commissioner of Patents and Trademarks